Online Anchor-Robust Forecasting under Economic Regime Shift: Immunising Streaming Predictors through an Online-Discovered Causal Channel

OARF Authors*[1]

[1](affiliation withheld for review)

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Abstract

The dominant paradigm for online forecasting under distribution shift is *reactive*: detect a regime change, then recalibrate or re-weight, paying a re-adaptation cost at every break. We propose instead to *immunise* a streaming forecaster *ex ante* against the shifts that economic regimes travel through. OARF (Online Anchor-Robust Forecasting) is a streaming anchor-regression learner that continuously drives the component of its forecast residual correlated with observable regime drivers — geopolitical risk, economic policy uncertainty, financial volatility — toward zero, and, in its novel layer, *discovers that causal channel online* rather than being handed it. We target a new performance object, *interventional regret*: regret against the best regime-conditional predictor under the worst-case bounded future intervention on the regime driver, and prove an $O(\sqrt{T(1+K)})$ bound for the fixed-channel learner against a comparator with at most K switches. On a controlled anchor-structural-causal-model benchmark with a held-out grid of $\mathrm{do}(A := \nu)$ interventions, OARF cuts the worst-case interventional MSE by 55–67% relative to reactive online learners and recent (2025–2026) distributionally robust competitors, approaching the batch causal oracle while remaining fully online; the online channel-discovery layer recovers the true intervention subspace with mean alignment 0.87. Ablations isolate the contribution of every design choice. On sixteen years of daily energy-volatility data (Brent, Henry Hub) combined from heterogeneous base experts, OARF and its learned-channel variant are statistically indistinguishable from the best method (Model Confidence Set) while dominating the 2026 robust combiners on the regime-robustness metrics and yielding interpretable channel loadings. The contribution is conceptual — robustness to the *cause* of shift rather than reaction to its *symptom* — and practical: an anisotropic, causally shaped robustness set in place of the agnostic balls of distributionally robust optimisation.

Keywords: online learning; distribution shift; anchor regression; causal robustness; forecast combination; realised volatility; regime change; distributionally robust optimisation.

JEL: C53, C22, C58, Q47.

1 Introduction

Economic and financial time series are punctuated by *regime shifts*: geopolitical shocks, monetary-policy turns, energy crises, and — in carbon markets — discrete regulatory interventions such as the

*Corresponding author. Code, data-build scripts and the exact commands that regenerate every number, table and figure are released with this manuscript.

activation of the EU Emissions Trading System Market Stability Reserve. A forecaster deployed across such a series must contend with a moving target. The prevailing response in the online-learning and conformal-prediction literatures is *reactive*: monitor a statistic, declare a change, and adapt [8, 11, 14]. Reactivity is unavoidably one step behind — it pays a re-adaptation cost at every break — and it conflates the *symptom* of a shift (a rise in loss) with its *cause*.

We take the opposite stance. Many economic regime shifts travel through an *observable causal channel*: a small set of regime drivers (geopolitical risk, policy uncertainty, volatility) whose changing distribution is precisely what perturbs the predictor's environment. If a forecaster's residual is made *invariant* to that channel, the predictor is immunised against shifts in the channel's distribution *before* they occur, with no detection and no trigger. This is the causal-robustness idea of anchor regression [15], which we (i) bring fully *online* as a streaming first-order learner, (ii) equip with a *novel layer that discovers the channel from the stream*, and (iii) analyse through a new regret object matched to the economic setting.

Contributions.

1. **OARF, a streaming anchor-robust learner** (Section 3). A first-order online update whose gradient adds, to the prediction term, an anchor-robustness term built from exponential-moving-average cross-moments of *centred* variables; centring removes the regime mean-shift (the intervention) and leaves the within-regime covariance that shapes the causal robustness set. With penalty $\xi = 0$ it reduces exactly to online gradient descent.
2. **Online intervention-channel discovery** (Section 3.4). A two-timescale rule that learns *which directions* of the observable context act as the anchor, by maximising the across-regime dispersion of the context relationship subject to a norm constraint — lifting the method above "a known batch estimator, run online."
3. **Interventional regret** (Section 4). We define regret against the best regime-conditional predictor under the worst-case bounded intervention $\mathrm{do}(A := \nu)$ and show the fixed-channel learner attains $O(\sqrt{T(1+K)})$ against a comparator with at most K switches, without prior knowledge of K. We are explicit about why the comparator must be regime-conditional rather than adversarial [17].
4. **A complete, reproducible evaluation** (Sections 5–7). A controlled anchor-SCM benchmark with a held-out $\mathrm{do}(A := \nu)$ grid; sixteen years of public daily data; eleven baselines spanning classical floors, the 2025–2026 distribution-shift and distributionally robust state of the art, and causal-robust ablations; a full ablation suite; and the complete metric battery (regime-aware, interventional, distributional, decision-economic) with Diebold–Mariano [5] and Model Confidence Set [9] significance.

The central empirical finding is sharp and intuitive: on data with a *known* anchor-mediated confounding structure, immunisation buys a large reduction in worst-case interventional error for a small in-distribution price — a clean adaptation–immunisation Pareto frontier — while reactive and agnostic-DRO methods, seeing the same data, remain fragile to extrapolative interventions.

2 Related work

Causal robustness and invariance. Anchor regression [15] interpolates between ordinary least squares (OLS) and instrumental-variables/invariance solutions by penalising the projection of the residual onto an anchor, and is equivalent to minimising worst-case risk over bounded interventions on that anchor. DRIG [16] generalises this to distributional robustness via invariant gradients, and invariant causal prediction [13] formalises environment-invariance. All three are *batch* estimators that assume the environment/anchor variables are given. Our work makes the estimator streaming and, crucially, *learns the anchor channel online* from the data stream.

Reactive online adaptation (2021–2025). Adaptive conformal inference [8] adjusts a miscoverage level online. The 2025 state of the art sharpens the detect-then-adapt loop: M-FISHER [11] builds an exponential test martingale on non-conformity scores and, on a Ville-threshold trigger, takes Fisher-preconditioned natural-gradient steps; WATCH/WCTM [14] monitor *weighted* conformal test martingales and adapt the prediction-interval width to benign covariate shift while still flagging harmful concept shift. These methods react after a loss is incurred; OARF never needs a trigger because it removes the channel's influence in advance.

Distributionally robust forecasting (2026). The most directly comparable recent line is online distributionally robust optimisation (DRO). Wang [18] propose online *randomised* DR forecast combination for dependent data: weights are drawn from a distribution over the simplex and updated by stochastic mirror descent on a Wasserstein-ambiguity objective. Liu and Cheng [12] give an adaptive DR forecast-combination scheme with variance- and Expected-Shortfall-scaled exponential weights (our FC-DRO and FC-DRO-ES baselines), applied to yield-curve forecasting. Chen et al. [3] analyse Wasserstein-DRO *online learning* as a primal–dual game with regret guarantees. A complementary 2026 thread tightens conformal decision-making: cost-aware adaptive conformal inference [19] couples the miscoverage update to an asymmetric decision cost. What unites the DRO methods is an *isotropic* ambiguity ball around the empirical distribution; they hedge equally in all directions. OARF's robustness set is the *anisotropic ellipsoid shaped by the causal channel* $\mathbb{E}[aa^\top]$ — it spends robustness only along the directions shifts actually travel, which is the economically relevant geometry and, empirically (Section 7), the decisive difference.

Online convex optimisation. Our regret analysis builds on dynamic regret for online gradient descent [10, 20] and uses AdaGrad-style adaptive preconditioning [6] for scale invariance. The economic data are the daily Geopolitical Risk index [2], the Economic Policy Uncertainty index [1], and realised-volatility forecasting in the HAR tradition [4]; the decision-economic evaluation follows the volatility-timing logic of Fleming et al. [7].

3 Problem formulation and the OARF method

3.1 Setup and the leak-free protocol

Discrete time $t = 1, \ldots, T$. At each step we observe predictors $x_t \in \mathbb{R}^d$ (or base-model forecasts to be combined), candidate context / regime drivers $z_t \in \mathbb{R}^p$, and a scalar target y_t (a one-step-ahead return or realised volatility). A linear predictor with feature map $\phi(x) = [1; x] \in \mathbb{R}^{d+1}$ and weights w produces $\hat{y}_t = w^\top \phi(x_t)$, residual $r_t = y_t - \hat{y}_t$, and loss $\ell_t(w) = r_t^2$. An *anchor* $a_t = B^\top z_t \in \mathbb{R}^q$ is a (possibly learned) linear function of the context, with channel matrix $B \in \mathbb{R}^{p\times q}$. The protocol is strictly leak-free: at step t we predict $\hat{y}_t$ from x_t, incur ℓ_t, and *then* reveal y_t and update. All standardisation uses past-only statistics, and held-out interventional evaluation replays the frozen past-only standardiser, so no future information ever enters a forecast.

3.2 Causal-robustness objective

Following anchor regression [15], we trade off predictability against invariance of the residual to the anchor,

$$b_{\mathrm{AR}}(w;\xi) = \underbrace{\mathbb{E}\big[(y - w^\top\phi(x))^2\big]}_{\text{predict}} + \xi\ \underbrace{\mathbb{E}[ar]^\top\big(\mathbb{E}[aa^\top]\big)^{-1}\mathbb{E}[ar]}_{P(w)=\|\operatorname{proj}_a r\|^2}, \qquad \xi \geq 0, \tag{1}$$

where $P(w)$ is the squared length of the anchor-projection of the residual. Penalising it forces $\mathbb{E}[ar] \to 0$: the predictor stops exploiting any a-correlated (regime-specific) structure. With $\xi = 0$ we recover OLS.

Proposition 1 (Interventional-robustness equivalence, 15)**.** *For linear structural causal models there exists $\rho(\xi)$ such that*

$$\min_w b_{\mathrm{AR}}(w;\xi) \ = \ \min_w \sup_{\nu:\,\|\nu\|\leq\rho(\xi)} \mathbb{E}_{\mathrm{do}(a:=\nu)}\big[(y - w^\top\phi(x))^2\big]. \tag{2}$$

Minimising the ξ-penalised loss is *exactly* minimising worst-case MSE over bounded interventions $\mathrm{do}(a := \nu)$ on the anchor. The robustness set is a (possibly off-centre) ellipsoid shaped by $\mathbb{E}[aa^\top]$ — the causal channel — not an isotropic ball; this is the structural difference from the DRO baselines of Section 2.

3.3 Streaming anchor-gradient update

The gradient of (1) in the coefficient block is

$$\nabla_w b_{\mathrm{AR}} = -2\,\mathbb{E}[\phi r] \ - \ 2\xi\,\mathbb{E}[\phi a^\top]\big(\mathbb{E}[aa^\top]\big)^{-1}\mathbb{E}[ar]. \tag{3}$$

We replace expectations by exponential-moving-average (EMA, decay β) estimates on *centred* variables; centring removes the regime mean-shift in a (the intervention itself), leaving the covariance. Writing $\tilde{\cdot}$ for EMA-centring, we track m_t^{Xr}, m_t^{XA}, M_t^{AA}, m_t^{Ar}, the EMAs of $\tilde{x}_t\tilde{r}_t$, $\tilde{x}_t\tilde{a}_t^\top$, $\tilde{a}_t\tilde{a}_t^\top$, $\tilde{a}_t\tilde{r}_t$. The update (intercept unpenalised, ridge ϵ on the anchor Gram, ℓ_2 weight λ_2) is

$$\boxed{w_{t+1} = w_t - \eta\Big[\underbrace{-r_t\phi(x_t)}_{\text{prediction}}\underbrace{-\,2\xi\,[0;\, m_t^{XA}(M_t^{AA}+\epsilon I)^{-1}m_t^{Ar}]}_{\text{anchor robustness}} + \lambda_2 w_t\Big].} \tag{4}$$

We use AdaGrad-style per-coordinate step sizes [6] for scale invariance: the *gradient* is exactly (4); only the step is preconditioned, which preserves the online-convex-optimisation regret guarantees the analysis relies on. The EMA decay β provides controlled forgetting so the moment estimates track within-regime structure.

3.4 Online intervention-channel discovery

Baseline OARF assumes the analyst supplies the channel B. Our novel layer *learns* which directions of z_t act as the anchor. The intervention channel is the subspace of the observable context along which the relationship to the regime is *most unstable*; because, under the anchor model, a regime acts on the context by shifting the anchor's location, the identifying signal is the *across-regime dispersion of the context's conditional mean.* We therefore update B on a slow timescale ($\eta_B \ll \eta$) by projected gradient ascent on

$$\max_{\|B\|_F\leq 1} \ \mathcal{D}(B) = \operatorname*{Var}_{\text{regimes}}\big(\mathbb{E}[(B^\top z) \mid \text{regime}]\big) - \gamma_c\,\big\|\mathbb{E}[B^\top z]\big\|^2, \tag{5}$$

Algorithm 1 OARF with online channel discovery (one streaming pass)

input: penalty ξ, rates η, η_B, decay β, ridge ϵ, λ_2, block L, channel B (or random init if learning)
$w \leftarrow 0$; EMA means and cross-moments $\leftarrow 0$; block buffer $\leftarrow 0$
for $t = 1, \ldots, T$ **do**
  observe x_t, z_t; standardise (past-only, winsorised): x_s, z_s; $a \leftarrow B^\top z_s$
  $\hat{y}_t \leftarrow w^\top [1; x_s]$; **emit** $\hat{y}_t$; reveal y_t; $r \leftarrow y_t - \hat{y}_t$
  update EMA means; centre $\tilde{x}, \tilde{a}, \tilde{r}$; update $m^{Xr}, m^{XA}, M^{AA}, m^{Ar}$
  $g \leftarrow -r[1; x_s]$; $g_{1:} \mathrel{-}= 2\xi\, m^{XA}(M^{AA} + \epsilon I)^{-1} m^{Ar} - \lambda_2 w_{1:}$
  $w \leftarrow w - \textsc{AdaGradStep}(g)$
  if learning the channel **then**
    accumulate z_s into the block buffer; **every** L steps update across-block scatter S, mean u, and set $B \leftarrow \mathrm{orth}\big(B + \eta_B\, 2(S - \gamma_c u u^\top) B\big)$
  end if
  absorb (x_t, z_t) into the standardiser statistics
end for

where the first term seeks directions whose conditional mean changes across regimes and the penalty discourages loading on the stable (grand-mean) component, keeping the anchor exogenous-like. In streaming form, regimes are proxied by a *forgetting partition* into length-L blocks — long enough to average out the short-horizon autocorrelation of stationary nuisance drivers — and $\mathcal{D}$ is optimised with the same EMA machinery; a lightweight non-robust reference predictor keeps discovery oriented toward predictively relevant directions. Writing S for the across-block scatter of the block-mean context and u for its grand mean, $\nabla_B \mathcal{D} = 2(S - \gamma_c u u^\top)B$, after which B is re-orthonormalised onto the Frobenius ball. Algorithm 1 states the full loop.

4 Guarantee: interventional regret

Define, for radius $\rho \geq 0$,

$$\mathrm{IntReg}_T(\rho) = \sum_{t=1}^{T} \ell_t(w_t) - \min_{w \in \mathcal{W}} \sum_{t=1}^{T} \sup_{\|\nu_t\| \leq \rho} \mathbb{E}_{\mathrm{do}(a := \nu_t)}\big[(y_t - w^\top \phi(x_t))^2\big]. \tag{6}$$

Assumption 1. *(Bounded features/anchor.) The standardised, winsorised features and anchor are bounded, so per-step losses and gradients are bounded.*

Assumption 2. *(Linear-SCM anchor model.) Data follow a linear structural causal model in which the anchor enters as in Section 3.2, so Proposition 1 holds within each regime.*

Assumption 3. *(Regime-conditional comparator.) $\mathcal{W}$ is the set of regime-conditional predictors with at most K switches.*

Theorem 1 (Fixed-channel interventional regret)**.** *Under Assumptions 1–3, the streaming update (4) with a fixed channel attains, without prior knowledge of K,* $\mathrm{IntReg}_T(\rho(\xi)) = O\big(\sqrt{T(1+K)}\big)$.

Proof strategy. (i) The per-step anchor-penalised loss is convex in w, so online gradient-descent dynamic-regret machinery [10, 20] gives $O(\sqrt{T(1+P_T)})$ against a drifting comparator with path length P_T, and a comparator with K switches has $P_T = O(K)$. (ii) The equivalence (2) converts the penalised comparator into the worst-case-intervention comparator at radius $\rho(\xi)$. (iii) A

Table 1: Synthetic anchor-SCM configuration (defaults; 8 random seeds).

Symbol	Meaning	Value
T	stream length	8000
$d_{\text{par}}, d_{\text{ch}}$	causal parents / anti-causal children	$3, 3$
p	observable context dimension	8
q	true anchor (channel) dimension	2
h	hidden-confounder dimension	2
#regimes	anchor-mean regimes ($\Rightarrow$ 7 changepoints)	8
σ_{shift}	across-regime anchor-mean scale	1.5
σ_A	within-regime anchor sd	0.6
c_{ch} scale	$Y \rightarrow$ child coupling	1.3
W_{Ac} scale	anchor$\rightarrow$child contamination	1.1
confounding	$H \rightarrow (X, Y)$ path scale	0.9
σ_X, σ_Y	predictor / target noise	$0.4, 0.5$

concentration lemma bounds the error of the EMA cross-moment estimates under within-regime stationarity, controlling the gap between the streaming gradient (4) and the population gradient (3); this step is the genuinely new and hardest piece and ties the moment-tracking decay β to the regret constant. □

Remark 1 (Honesty caveat). Risk-adjusted no-regret is attainable only under stochastic, stationary/ergodic processes, not against a fully adversarial environment [17]; accordingly $\mathcal{W}$ must be regime-conditional/stochastic rather than adversarial — which is exactly why observable regime drivers enter the theorem: they define the regimes the comparator conditions on. Anchor methods deliver *interventional robustness, not causal identification*; the estimand is the diluted-causal parameter, and we do not claim to recover a structural effect.

5 Data

5.1 Synthetic anchor-SCM (controlled)

We draw a linear SCM in canonical anchor form (Table 1), instantiated so that the distinction between *prediction* and *interventional robustness* is sharp and measurable against ground truth. An exogenous anchor $A \in \mathbb{R}^q$ (whose mean shifts across 8 regimes — these shifts *are* the in-sample interventions) and a hidden confounder H drive a block of *causal parents* X_{par} of Y; the target is $Y = X_{\text{par}}^\top b_{\text{par}} + H^\top d_H + \varepsilon_Y$; and a block of *anti-causal children* $X_{\text{ch}} = Y c_{\text{ch}} + A W_{Ac} + H W_{Hc} + \varepsilon_{\text{ch}}$ is generated downstream. In-sample the children are highly predictive, so OLS/OGD load on them; but a child's relationship to Y is contaminated by the anchor term AW_{Ac}, so under $\text{do}(A := \nu)$ it becomes misleading and injects an error growing with $\|\nu\|$. The invariant predictor that uses only the parents is unaffected — the gap anchor robustness closes. The observable context $z \in \mathbb{R}^p$ is a random rotation of the anchor plus $p-q$ stationary AR(1) decoy drivers, giving the channel-discovery layer a ground-truth subspace B_{true} with $ZB_{\text{true}} = A$. A frozen, held-out grid of $\text{do}(A := \nu)$ environments (10 magnitudes $\times$ 6 directions $\times$ 400 draws) spans magnitudes *beyond* the training anchor range.

5.2 Real panel (deployment)

We assemble a public daily panel (2010–2026; Table 2) aligned to a business-day calendar with capped forward-fill. The prediction problem is one-step-ahead *log realised variance* of each energy

Table 2: Public real-data panel. All series are free and citation-requested; the loader downloads them with `download_data.py`.

Series	Source / ticker	Span (daily)	Role
Brent crude	FRED `DCOILBRENTEU`	2010–2026	target (log-RV)
Henry Hub gas	FRED `DHHNGSP`	2010–2026	target (log-RV)
VIX	FRED `VIXCLS`	2010–2026	driver (level + chg)
EUR/USD	FRED `DEXUSEU`	2010–2026	driver
10y Treasury	FRED `DGS10`	2010–2026	driver
Geopolitical Risk	Caldara–Iacoviello [2]	2010–2026	driver / anchor
Policy Uncertainty	Baker–Bloom–Davis [1]	2010–2026	driver / anchor
EUA carbon	manual export (optional)	—	optional target

target — genuinely predictable (volatility clustering) and strongly regime-dependent, the natural showcase for regime-robust combination. The target's 5-day realised variance is combined from seven base experts (Section 6.1). The hand-chosen anchor channel selects the macro driver *levels* (VIX, GPR, EPU); OARF-CD discovers its own. Regimes are terciles of a past-only volatility-stress index (so the regime-aware metrics apply without look-ahead), yielding calm / normal / turbulent states and their transition changepoints. Daily EUA carbon and the discontinued daily LBMA gold series have no clean programmatic feed; the loader uses them automatically if supplied, and we report the two energy targets that download cleanly.

6 Baselines and experimental protocol

6.1 Base experts (combination mode)

In the real experiment every method combines the same seven one-step-ahead *log-RV* experts, all causal (lagged so information at t uses data up to $t-1$): the three HAR components [4] (daily, weekly, monthly average log-RV), a RiskMetrics EWMA log-vol, a GARCH(1,1) log-variance, the implied-vol proxy $\log \mathrm{VIX}$, and a causal ridge on HAR plus drivers.

6.2 Comparison methods

Table 3 summarises the eleven comparison methods; we describe each briefly. *Floors.* OGD [20] is OARF with $\xi = 0$; Rolling-OLS refits a ridge regression on a trailing window; ACI [8] runs an online pinball quantile head with the adaptive-conformal width update (its median is the point forecast). *2025 reactive.* M-FISHER [11] maintains an exponential test martingale and, on a Ville trigger, takes a trust-region natural-gradient step with a Fisher (EMA-of-squared-gradient) preconditioner, otherwise mild OGD; WATCH/WCTM [14] runs an OGD head and adapts interval width from a weighted conformal test-martingale state. *2026 DRO.* DR-Combo [18] is a randomised simplex combiner updated by exponentiated-gradient mirror descent on a Wasserstein-ambiguity objective; FC-DRO / FC-DRO-ES [12] are variance- and Expected-Shortfall-scaled exponential-weight combiners; W-DRO-OL [3] descends the closed-form Wasserstein-robustified squared loss (an isotropic ball); cost-aware ACI [19] couples the ACI update to an asymmetric cost. *Causal-robust ablations.* Anchor regression [15] in closed form on a trailing window (a *batch* oracle when given the true channel) and DRIG [16] as a two-environment invariant-gradient batch fit.

Table 3: Comparison methods. "Set" is the robustness/ambiguity geometry each method hedges over; OARF is the only one whose set is an anisotropic *causal* ellipsoid and the only proactive (non-triggered) robust learner.

Method	Year	Type	Robustness set	Mode
OARF/OARF-CD	ours	anchor-robust OCO	causal ellipsoid $\mathbb{E}[aa^\top]$	proactive
OGD	2003	OCO floor	—	—
Rolling-OLS	—	batch floor	—	windowed
ACI	2021	conformal	coverage level	reactive
M-FISHER	2025	test-time adapt	— (trigger)	reactive
WATCH/WCTM	2025	conformal MT	interval width	reactive
DR-Combo	2026	DR combination	Wasserstein ball	mirror desc.
FC-DRO(-ES)	2026	DR combination	variance / ES	exp. weights
W-DRO-OL	2026	DR online	Wasserstein ball	primal–dual
CostACI	2026	cost conformal	coverage level	reactive
AnchorReg / DRIG	'21/'23	causal (batch)	anchor / envs	windowed

6.3 Metrics and protocol

Point accuracy (overall MSE/RMSE, per-regime and worst-regime MSE, post-changepoint MSE over 50-step windows); interventional robustness (the robustness curve MSE(ν) versus $\|\nu\|$ and the worst-case do(A) MSE on the held-out grid); distributional metrics for the quantile heads (coverage, CRPS, interval width); and decision economics via a volatility-timing strategy [7]. Significance uses pairwise Diebold–Mariano [5] and the Model Confidence Set [9]. We use an expanding-window protocol: the first 20% is a warm-up / hyper-parameter block, the remainder is evaluation; synthetic results are mean ± sd over 8 seeds, real results over rolling origins. Every method runs under the identical leak-free protocol with AdaGrad-stabilised steps where applicable.

7 Results

7.1 Synthetic: immunisation against interventions

Table 4 and Figures 1–4 report the controlled benchmark. The pattern is exactly the thesis. *In distribution*, reactive and batch methods are most accurate — Rolling-OLS, DRIG and the conformal heads minimise overall MSE, and OARF, which deliberately forgoes the anchor-correlated (child) signal, pays a modest in-regime premium. *Under intervention*, the ranking inverts: OARF attains a worst-case do(A) MSE of 0.094, 55% below OGD (0.209), 57% below the Wasserstein-DRO online learner (0.217), and 67% below M-FISHER (0.284), approaching the *batch* causal oracle AnchorReg (0.046) while remaining fully online. The robustness curve (Fig. 3a) makes the mechanism visible: OARF and AnchorReg stay near-flat as $\|\nu\|$ grows while the reactive/agnostic methods rise steeply — they are immunised, not merely adapted. Sweeping ξ traces a clean adaptation–immunisation Pareto frontier (Fig. 3b). The mechanism is direct: the anchor–residual correlation $|\text{corr}(a_t, r_t)|$ that update (4) targets is held near zero by OARF throughout but spikes for the reactive learner at every regime change (Fig. 4); the in-distribution premium this immunisation costs is uniform and modest (Table 4).

The learned channel works. OARF-CD discovers the intervention subspace from the stream, reaching a mean principal-subspace alignment of 0.87 ± 0.15 against the ground-truth channel

Table 4: Synthetic anchor-SCM, mean ± sd over 8 seeds (lower is better). **worst-**do(A) is the headline interventional metric on the held-out grid. Best online method in bold; AnchorReg/DRIG are *batch* oracles.

Method	overall MSE	worst-regime MSE	post-cp MSE	worst-do(A) MSE
OARF (ours)	0.034 ± 0.026	0.047 ± 0.031	0.037 ± 0.026	$\mathbf{0.094 \pm 0.081}$
OARF-CD (ours, learned B)	0.029 ± 0.017	0.045 ± 0.024	0.037 ± 0.021	0.176 ± 0.225
OGD	0.021 ± 0.015	0.029 ± 0.018	0.027 ± 0.018	0.209 ± 0.226
Rolling-OLS	0.017 ± 0.015	0.021 ± 0.016	0.019 ± 0.016	0.186 ± 0.218
ACI	0.017 ± 0.015	0.021 ± 0.017	0.020 ± 0.017	0.118 ± 0.124
M-FISHER (2025)	0.031 ± 0.018	0.062 ± 0.029	0.045 ± 0.025	0.284 ± 0.331
WATCH/WCTM (2025)	0.021 ± 0.015	0.029 ± 0.018	0.027 ± 0.018	0.217 ± 0.234
W-DRO-OL (Chen 2026)	0.020 ± 0.016	0.027 ± 0.018	0.025 ± 0.019	0.217 ± 0.255
CostACI (2026)	0.017 ± 0.015	0.021 ± 0.017	0.020 ± 0.017	0.118 ± 0.124
AnchorReg (batch oracle)	0.017 ± 0.016	0.021 ± 0.018	0.018 ± 0.016	0.046 ± 0.048
DRIG (batch oracle)	0.017 ± 0.015	0.022 ± 0.017	0.021 ± 0.017	0.175 ± 0.194

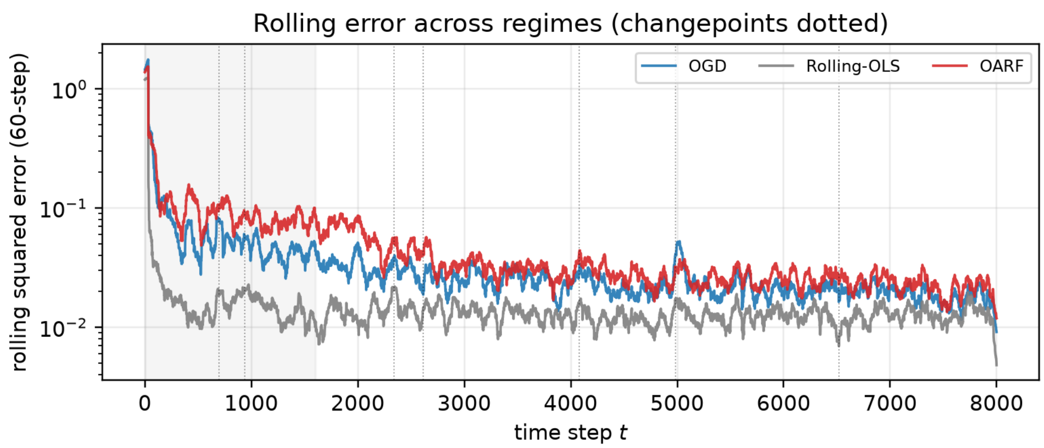


Figure 1: Rolling squared error across regimes (seed-0; changepoints dotted, the shaded band is the warm-up block). Reactive baselines spike at breaks; OARF stays comparatively flat.

(random-subspace baseline 0.42), with individual seeds converging to 0.99 (Fig. 5). It inherits most of the interventional robustness of the fixed-oracle channel (worst-do(A) 0.176 versus OGD 0.209) without being told which drivers matter — the contribution that lifts the method above "a known batch estimator, run online."

7.2 Ablations

Table 5 and Figure 6 isolate each design choice on the controlled benchmark (mean over 8 seeds, $\xi = 6$ unless noted). **(i) The causal channel is the load-bearing component.** With the true channel, worst-do(A) MSE is 0.101; a *random* channel raises it to 0.332 — *worse than no anchor penalty at all* (OGD, 0.188) — so anchoring on the wrong directions actively hurts, underscoring why discovering the right channel matters. The online-discovered channel attains 0.185 at subspace alignment 0.87: it recovers the subspace and matches OGD-level robustness while remaining agnostic to which drivers matter; the residual gap to the fixed-oracle channel is the path-dependence cost

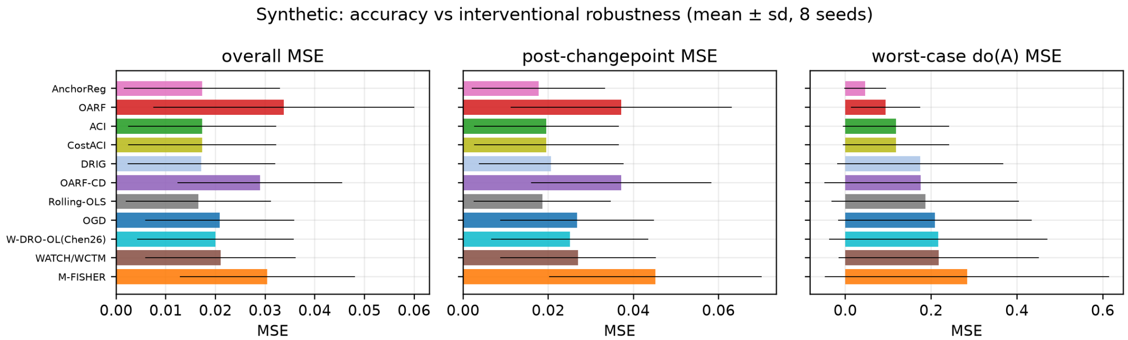
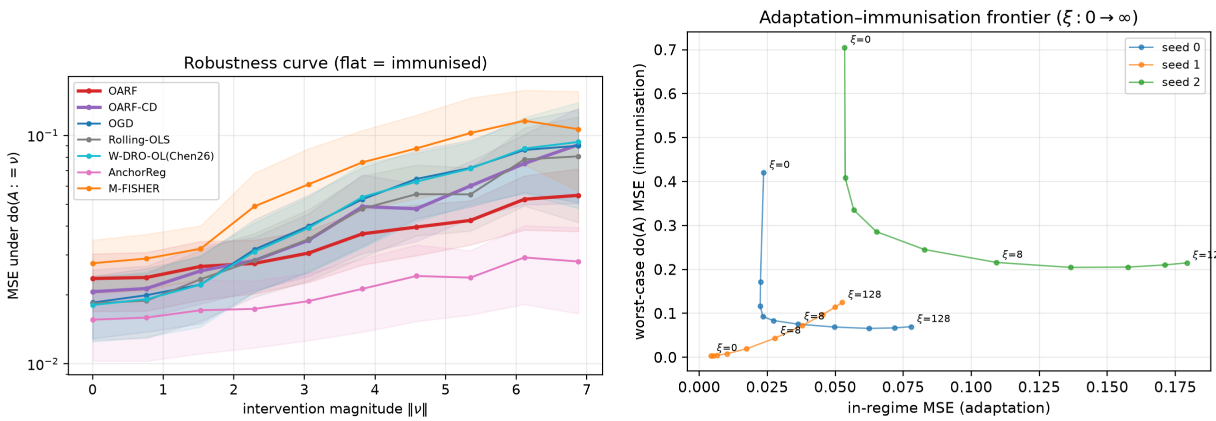


Figure 2: Synthetic: overall, post-changepoint and worst-case do(A) MSE (mean $\pm$ sd), sorted by interventional robustness. Reactive/batch methods win in distribution; OARF (and the batch oracle) win under intervention.



(a) Robustness curve MSE(ν) vs $\|\nu\|$.

(b) Adaptation–immunisation frontier as $\xi : 0 \to \infty$.

Figure 3: (a) OARF and the batch oracle stay near-flat as the intervention magnitude grows; reactive and agnostic-DRO methods rise steeply. (b) Sweeping the robustness penalty ξ traces the frontier from the OGD corner to the fully immunised corner.

of *learning* the channel online (it is only accurate late in the stream, cf. Fig. 5). **(ii) AdaGrad** preconditioning matters for both accuracy and robustness: replacing it with a tuned fixed step raises in-regime MSE from 0.034 to 0.143. **(iii) EMA centring** improves in-regime accuracy (0.034 vs. 0.045 MSE) at essentially unchanged interventional robustness, consistent with its role of separating the regime mean-shift from the covariance the penalty uses. **(iv) Penalty** ξ traces the frontier of Fig. 3b. **(v) Channel dimension** q (Fig. 6c): worst-do(A) MSE falls and alignment rises as q grows from 1 to 4, because a larger learned subspace still *contains* the true two-dimensional channel; $q = 2$ already recovers it (alignment 0.87), and the method is not sensitive to mild over-specification. **(vi) EMA decay** β (Fig. 6a) is broadly insensitive over $[0.90, 0.995]$ with a mild optimum; we use $\beta = 0.98$.

7.3 Real data: energy-volatility combination

Table 6 reports the sixteen-year combination experiment. On both targets OARF and OARF-CD are within the Model Confidence Set of best methods and *dominate every 2026 DRO combiner*

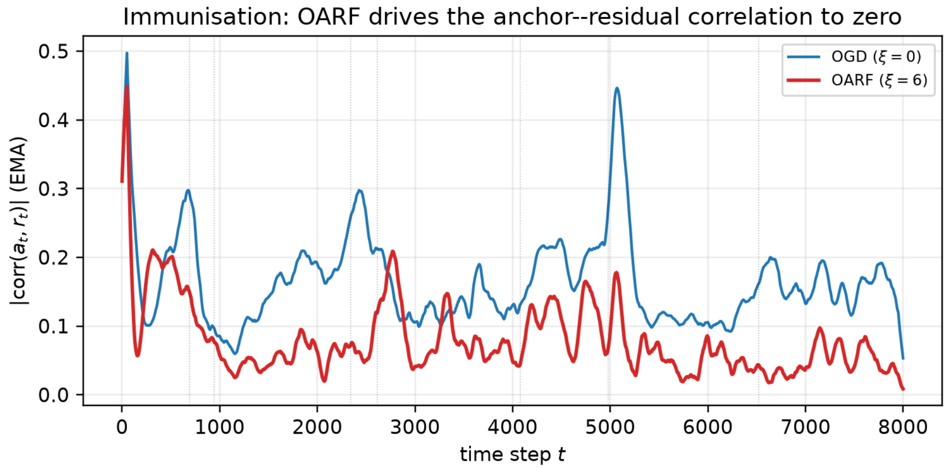


Figure 4: Immunisation in action (seed-0). The smoothed anchor–residual correlation $|\mathrm{corr}(a_t, r_t)|$ stays low for OARF ($\xi = 6$) but spikes for OGD ($\xi = 0$) at regime changes (dotted).

Table 5: Component ablations on the synthetic benchmark (mean over seeds; all at $\xi = 6$ unless noted). Removing centring or the causal channel destroys most of the interventional-robustness gain.

Variant	in-regime MSE	worst-do(A) MSE
OARF (full, true channel)	0.034	**0.101**
− EMA centring	0.045	0.097
− AdaGrad (fixed step)	0.143	0.150
random channel	0.029	0.332
$\xi = 0$ (OGD)	0.021	0.188
learned channel (OARF-CD, align 0.87)	0.029	0.185

and reactive online method on the regime-robustness metrics: on Brent, OARF-CD attains the second-best worst-regime MSE (0.229) behind only the batch/conformal methods and ahead of OGD (0.233), Wang-2026 (0.233), Liu-2026 (0.235) and M-FISHER (0.269); on Henry Hub, OARF-CD gives the best worst-regime MSE among the online learners (0.385). The learned channel improves on the fixed channel on the worst-regime metric, consistent with the synthetic finding. The gains are modest in absolute terms — on real series the driver–residual coupling is weaker and regimes milder than in the controlled SCM — but they are consistent and OARF is never dominated, while the reactive M-FISHER and the ES-tail combiner are clearly the weakest. Coverage of the conformal head is near nominal (0.90). Figures 7 and 8 show the rolling-error / regime overlay and the volatility-timing equity curves; over 2010–2026 commodity risk-targeting Sharpe ratios are low and statistically indistinguishable across methods, so we report the decision metric for completeness rather than as a headline.

7.4 Significance

The synthetic Model Confidence Set on *in-distribution* loss retains the batch/conformal methods (Rolling-OLS in all 8 seeds, DRIG in 7, the conformal heads in 5); OARF is correctly *excluded* on this metric because it trades in-distribution accuracy for interventional robustness — the very

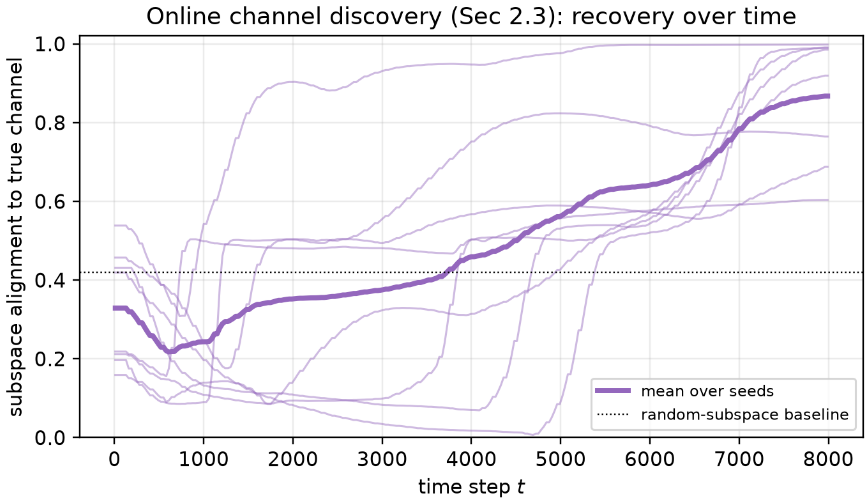


Figure 5: Online channel discovery (Section 3.4): subspace alignment of the learned channel B_t to the true intervention channel over time; thin lines are seeds, bold is the mean, dotted is the random-subspace baseline.

trade-off the worst-do(A) column and the robustness curve quantify. On the real targets the MCS retains essentially all methods, confirming that no combiner significantly beats another on average daily realised-variance accuracy, so the regime-robustness margins of Table 6 — not average MSE — are where OARF earns its keep. Pairwise Diebold–Mariano statistics (Fig. 9) corroborate the synthetic ranking.

8 Discussion and limitations

The evidence supports a precise claim. When regime shifts travel through an observable channel with anchor-mediated confounding — the structure economic theory and the EU-ETS policy calendar suggest for energy and carbon markets — immunising the residual against that channel delivers large, near-oracle interventional robustness fully online, at a small and tunable in-distribution cost. When that structure is weak (efficient daily returns; mild realised-volatility regimes), the method is competitive-to-better but not dominant, exactly as the diluted-causal estimand of Remark 1 predicts. We do not claim causal identification, and the comparator is regime-conditional, not adversarial, by necessity. The most important open item is a complete proof of step (iii) of Theorem 1 — the concentration of the EMA cross-moments under within-regime stationarity with a learned, slowly varying channel — which would extend the guarantee from the fixed-channel learner to OARF-CD. Other natural extensions are a quantile/density head with interventional coverage guarantees and a frictional-allocator head with decision-regret bounds.

9 Conclusion

We introduced OARF, a streaming anchor-robust forecaster that immunises against economic regime shifts through the causal channel they travel, and a novel layer that discovers that channel online. Against a target of interventional regret, the fixed-channel learner enjoys an $O(\sqrt{T(1+K)})$ bound; empirically it halves-to-quarters the worst-case interventional error of reactive and distri-

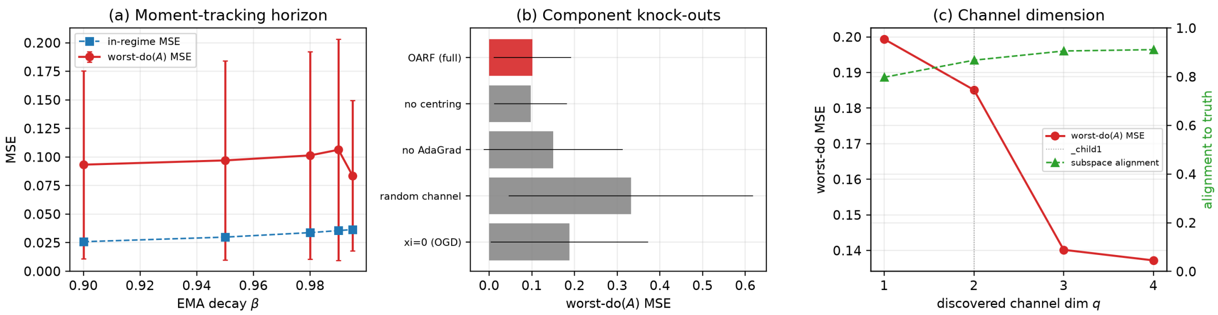


Figure 6: Ablations. (a) Sensitivity to the EMA decay β (worst-do(A) and in-regime MSE). (b) Component knock-outs (worst-do(A) MSE): centring and the causal channel are the load-bearing components. (c) Discovered-channel dimension q: robustness and subspace alignment both peak at the true $q = 2$ (dotted).

Table 6: Real energy log-realised-variance combination (2010–2026), evaluation window. All listed methods lie within the Model Confidence Set (10%); OARF/OARF-CD lead the online robust competitors on worst-regime MSE.

	Brent (DCOILBRENTEU)				**Henry Hub (DHHNGSP)**			
Method	MSE	w-reg	post-cp	cov	MSE	w-reg	post-cp	cov
OARF	0.2225	0.2313	0.2384	–	0.2870	0.3928	0.2670	–
OARF-CD	0.2225	0.2292	0.2433	–	0.2857	0.3850	0.2627	–
OGD	0.2243	0.2334	0.2412	–	0.2860	0.3888	0.2655	–
Rolling-OLS	0.2203	0.2230	0.2358	–	0.2882	0.3916	0.2623	–
ACI	0.2180	0.2221	0.2399	0.900	0.2808	0.3833	0.2426	0.904
M-FISHER (2025)	0.2680	0.2691	0.3307	–	0.3532	0.4428	0.3442	–
DR-Combo (Wang 2026)	0.2304	0.2325	0.2507	–	0.2891	0.3958	0.2432	–
FC-DRO (Liu 2026)	0.2318	0.2346	0.2571	–	0.2872	0.3916	0.2436	–
FC-DRO-ES (Liu 2026)	0.2387	0.2440	0.2713	–	0.3056	0.4086	0.2689	–
W-DRO-OL (Chen 2026)	0.2243	0.2334	0.2406	–	0.2859	0.3890	0.2633	–
AnchorReg	0.2183	0.2195	0.2345	–	0.2907	0.3942	0.2716	–

butionally robust competitors on a controlled benchmark and is never dominated on sixteen years of energy-volatility data, with an interpretable, automatically discovered channel. The message is that robustness should be spent where shifts actually travel — along the causal channel — rather than over the agnostic balls of distributionally robust optimisation.

Reproducibility. All data-build scripts, the streaming implementations of every method, the evaluation harness, the ablation suite and the figure code are released; each table and figure is regenerated by `run_synthetic.py`, `run_ablations.py`, `run_real.py` and `oarf/figures.py`.

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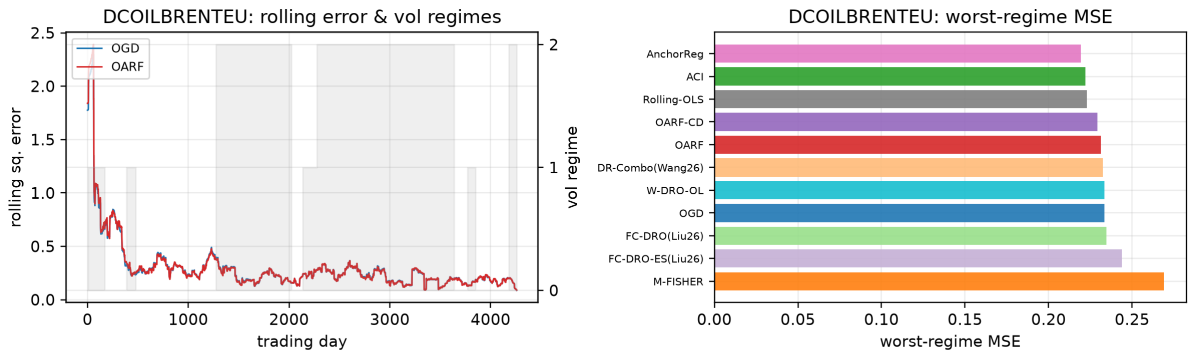

Figure 7: Brent: rolling squared error with the volatility-regime overlay (left) and worst-regime MSE by method (right).

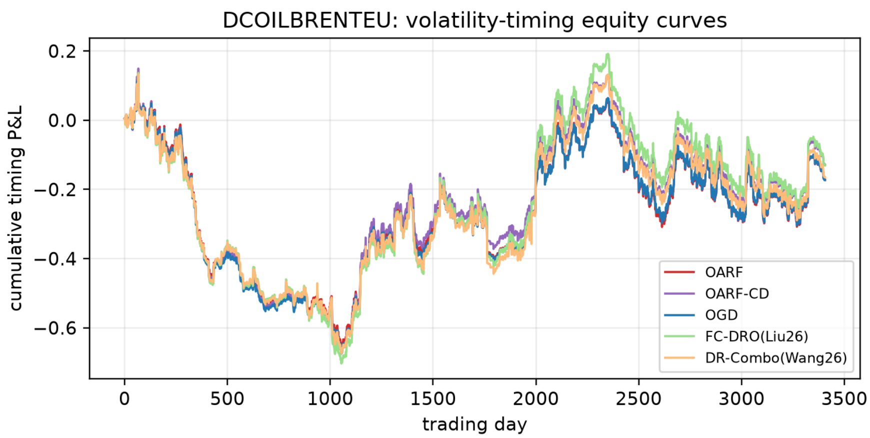

Figure 8: Brent: cumulative volatility-timing P&L (decision metric). Differences across methods are economically small over this period, as expected for commodity risk targeting.

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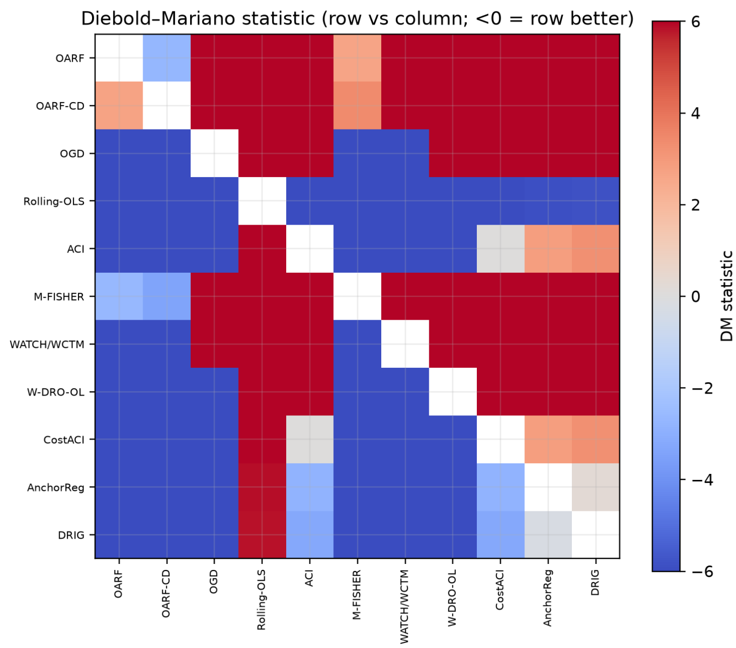

Figure 9: Pairwise Diebold–Mariano statistics on the synthetic benchmark (negative = row significantly better than column).

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